Gold Price Analytics Project

Complete submission with real dataset, notebook-based workflow, charts, model output, and final insights.

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Project Title: Gold Price Analytics using Real Monthly Historical Data

Problem Statement: I analyzed a real gold price dataset to understand how gold prices changed over time and what patterns are visible in long-term historical data. This project helps explain long-run price movement and monthly volatility in simple terms.

Dataset Description: The source dataset contains monthly gold prices in USD since 1833. For this project, the analytical dataset has 2319 rows and 8 columns after adding time-based and change-based features for better analysis.

1. Import Libraries

Code Cell - Import Libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import OneHotEncoder
from sklearn.impute import SimpleImputer
from sklearn.ensemble import RandomForestRegressor
```

All required libraries were imported in one code cell so the notebook can run cleanly from top to bottom.

2. Load Dataset and Prepare Columns

Code Cell - Load and Clean Data

```
df = pd.read_csv('gold_monthly.csv')
df['Date'] = pd.to_datetime(df['Date'])
df['Year'] = df['Date'].dt.year
df['Month_Num'] = df['Date'].dt.month
df['Month_Name'] = df['Date'].dt.strftime('%b')
df['Quarter'] = 'Q' + df['Date'].dt.quarter.astype(str)
df['Decade'] = (df['Year']//10*10).astype(str) + 's'
df['Price_Change'] = df['Price'].diff().fillna(0)
analysis_df = df[['Date','Price','Year','Month_Num','Month_Name','Quarter','Decade','Price_Change']]
analysis_df.head()
```

First 5 Rows

Date	Price	Year	Month_Num	Month_Name	Quarter	Decade	Price_Change
1833-01-01	18.93	1833	1	Jan	Q1	1830s	0.0
1833-02-01	18.93	1833	2	Feb	Q1	1830s	0.0
1833-03-01	18.93	1833	3	Mar	Q1	1830s	0.0
1833-04-01	18.93	1833	4	Apr	Q2	1830s	0.0
1833-05-01	18.93	1833	5	May	Q2	1830s	0.0

Dataset Shape

Rows	Columns
2319	8

Column Data Types

Column	Data Type
Date	datetime64[ns]
Price	float64
Year	int32
Month_Num	int32
Month_Name	object
Quarter	object
Decade	object
Price_Change	float64

The dataset includes one original numeric column (`Price`) and several derived columns such as `Month_Name`, `Quarter`, and `Decade`, which provide the required categorical information for analysis and prediction.

3. Data Cleaning

Missing Values Check

Column	Missing Values
Date	0
Price	0
Year	0
Month_Num	0
Month_Name	0
Quarter	0
Decade	0
Price_Change	0

No missing values remained in the final dataset. The first `Price_Change` value, which can be empty after a difference operation, was filled with 0, and duplicate rows were removed if present.

4. Statistical Analysis

Code Cell - Statistics

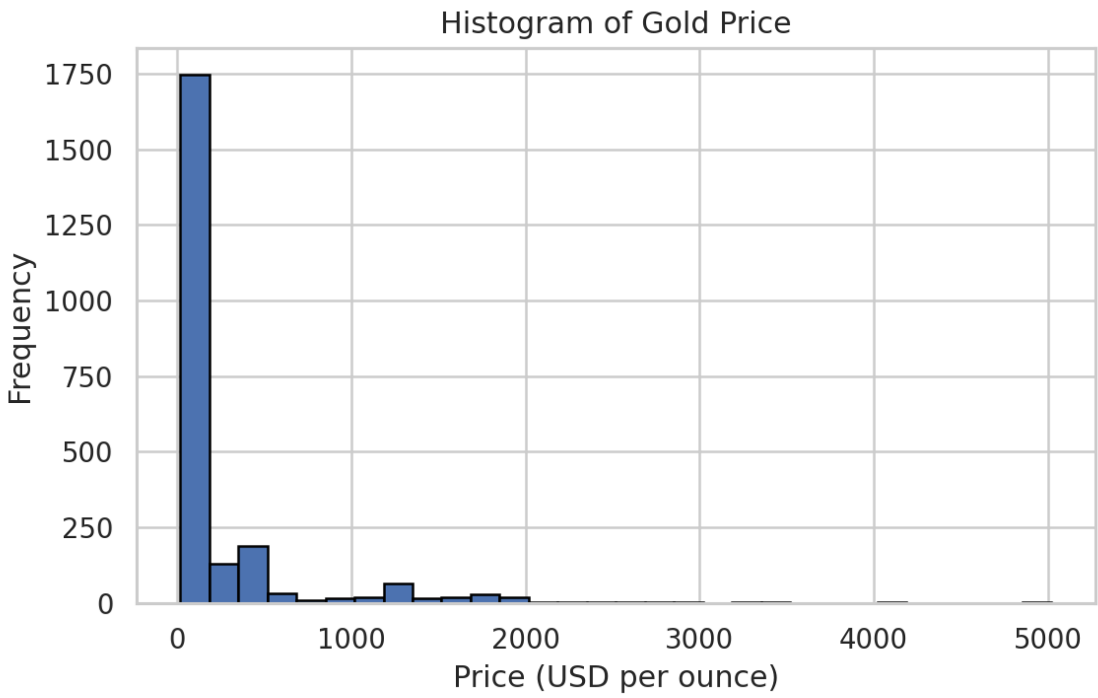
```
stats_df = pd.DataFrame({
    'Price': stats_for('Price'),
    'Price_Change': stats_for('Price_Change')
})
stats_df
```

Statistics Table

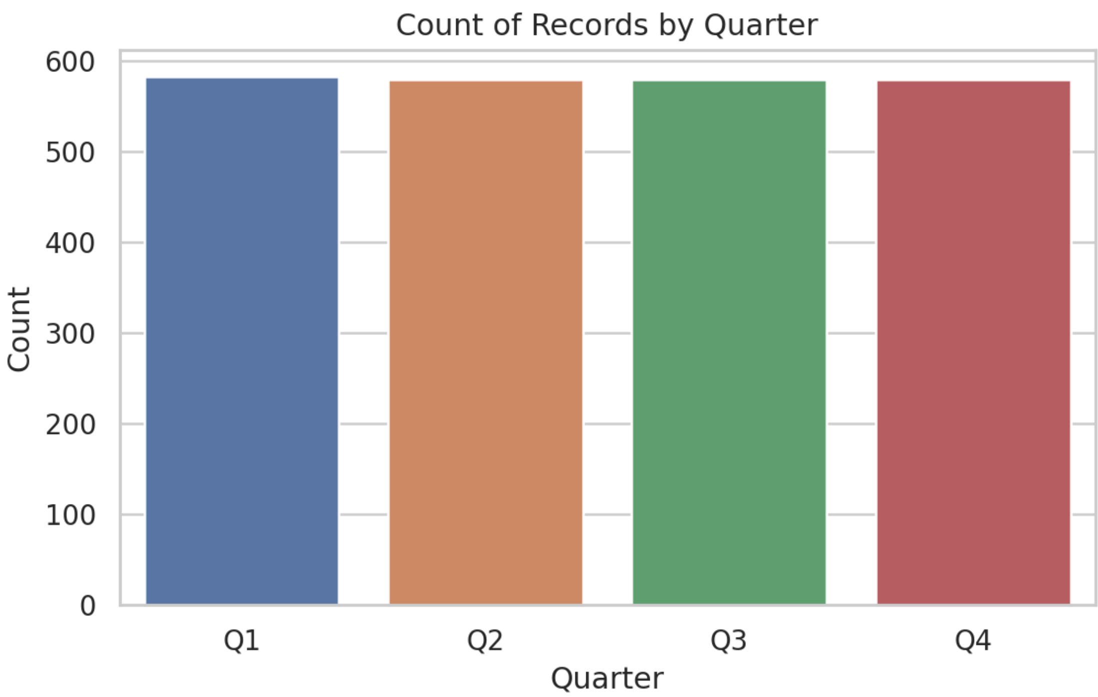
Statistic	Price	Price_Change
Mean	234.23	2.09
Median	20.68	0.0
Mode	18.93	0.0
Standard Deviation	517.57	24.91
Variance	267875.83	620.39
Range	5002.91	607.95
Mid-range	2518.52	139.55

The average gold price is 234.23, while the median is 20.68. Since the mean is much larger than the median, the distribution is influenced by high prices in recent years.

5. Visualizations

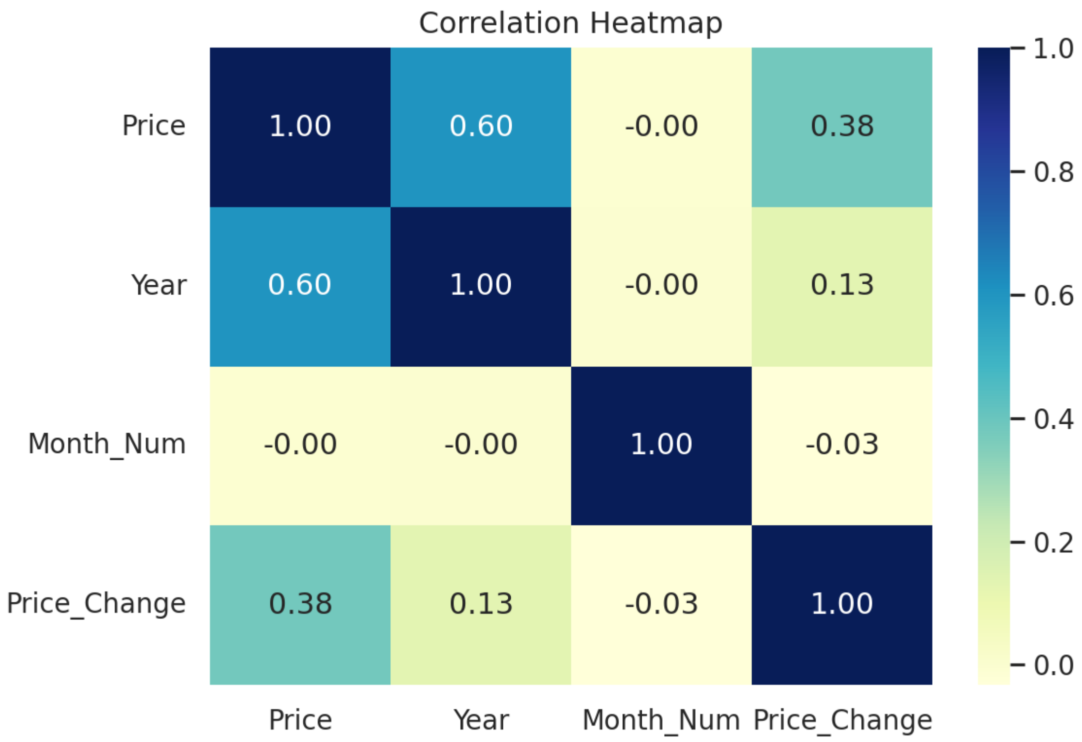


Histogram interpretation: The price distribution is right-skewed, meaning lower historical prices appear much more frequently than recent high prices.



Countplot interpretation: Each quarter has almost the same number of records because the dataset has one record per month across many years.

Boxplot interpretation: There are several outliers in monthly price change, which means some months had unusually sharp price increases or decreases.



Heatmap interpretation: `Price` has a strong positive relationship with `Year`, showing that gold generally increased in value across the long run.

6. Simple Prediction Model

Code Cell - Simple Prediction

```
X = analysis_df[['Year','Month_Num','Month_Name','Quarter','Decade','Price_Change']]
y = analysis_df['Price']

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

pipe.fit(X_train, y_train)
pred = pipe.predict(X_test)

r2_score(y_test, pred)
```

Model Metrics

Metric	Value
R-squared	0.9944
MAE	8.1
RMSE	33.05

The dataset was split into training and testing parts using an 80:20 ratio. A Random Forest Regressor was used after encoding the categorical columns. The model achieved an R-squared value of 0.9944, which means it explains most of the variation in the test data.

Simple English explanation: The model predicts gold price very well because the year, decade, and recent monthly change already carry strong information about the price level.

The recent 20-year trend chart also supports the model result because the data contains a clear long-term upward pattern.

7. Final Insights and Recommendations

Finding 1: The histogram shows a strong right-skew, so very high prices are mostly concentrated in recent years.

Finding 2: The statistics table shows a mean price of 234.23 and a median of 20.68, confirming that newer high prices pull the average upward.

Finding 3: The heatmap shows a strong positive correlation between `Price` and `Year`, which indicates long-term growth in gold prices.

Finding 4: The boxplot shows outliers in `Price_Change`, meaning some months were much more volatile than usual.

Recommendation 1: A non-technical user should focus on long-term trend rather than reacting to one-month jumps, because short-term changes can be noisy.

Recommendation 2: During months with high volatility, buyers and investors should be more careful because the boxplot shows that unusually large price changes do occur.

8. GitHub Link

Paste your GitHub repository link here after uploading the notebook, CSV files, PDF, and README.

GitHub Link: ________________________________